WOJCIECH CZYŻYK

EMBEDDED SYSTEMS ARCHITECT & FULL-STACK ENGINEER

📍 Jeżów Sudecki / Jelenia Góra, Lower Silesia, Poland
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Professional Profile: Software systems engineer with unique, 15+ years of experience at the intersection of hardware and software. Specialist in embedded systems (Embedded, Bare-Metal ARM), Hard Real-Time environments, industrial automation (BMS, Nice TTBus protocols), and modern application architecture (C++20, Perl, NodeJS, SQL). Combines profound low-level competencies with advanced knowledge of database engines, concurrency mechanisms, growth e-commerce analytics (SEO/Google Ads), and project management (Agile/Scrum). Holds a Class III glider pilot license (deep understanding of avionics, safety-critical workflows, and emergency systems/Aerospace).

🛠️ KEY COMPETENCIES (TECH STACK)

Embedded & Hardware	Bare-Metal ARM (Cortex-M, ARM7TDMI), FPGA programming (VHDL), 8051 microcontrollers (Assembler), signal processing & conversion (ADC/DAC), IR receivers & demodulators (TSOP series), analog circuit design & simulation (pSpice).
Reliability & Safety-Critical	Design of redundant hardware systems with hardware arbitration and voting blocks (TMR - Triple Modular Redundancy) compliant with Mission-Critical standards (SIL/ Aerospace/Automotive).
Protocols & Communication	Industrial reverse engineering of proprietary wireless and wired protocols (Nice TTBus, RC5, NEC, Sony, VoiceXML). Practical experience in radio frequency (RF) engineering and transmission lines on PCB impedance matching levels.
E-commerce & Growth Analytics	Search Engine Optimization (SEO), advanced user traffic and conversion analysis using Google Analytics, strategic budgeting and optimization of paid search campaigns (Google AdWords / Ads).
Software Development	C++20, C, Perl, NodeJS, HTML/CSS.
Databases & Concurrency	MySQL, SQLite – advanced transactional engines (ACID), structural locking mechanisms (Table/Row Locking) optimized for high concurrency, multi-threaded pipelines, and hard real-time execution.
Management & Tools	GitHub Enterprise, GitHub Projects (advanced data mapping, Kanban/Table layout automation), UML (class architecture design, Clean Architecture/MVC), ActiveNow, iFirma.

💼 PROFESSIONAL EXPERIENCE & ENGINEERING PROJECTS

Lead Embedded Systems Engineer – Commercial Measurement Devices **2010 – 2020**

Client: ZacharPLC (Karpacz, Poland) — 10 years of continuous product lifecycle development

- **Architecture of the "Chrono Zachar" Ecosystem:** Designed, programmed, and evolved an autonomous race-timing chronometer engineered for ultra-precise time capture in mass-participation and extreme sporting events (e.g., "3xŚnieżka = Mount Blanc" mountain ultra-marathon, 1/4-mile drag racing, ski slaloms in sub-zero alpine conditions, and TFA firefighter combat challenges).
- **Bare-Metal ARM Firmware & Input Capture Determinist:** Developed low-level firmware (C/Assembler) for 32-bit **NXP LPC2140** / **LPC2148** processors (ARM7TDMI-S core) directly manipulating hardware registers without HAL abstraction. Utilized hardware **Input Capture (CR)** registers to freeze clock timestamps on a pure silicon layer, entirely eliminating software interrupt latency errors.
- **Custom Memory Allocator & Object-Oriented Refactoring:** Executed a comprehensive 2-month architectural migration from procedural C to object-oriented C++. Overloaded the global new operator to enforce deterministic **Memory Pooling** within fixed RAM boundaries, completely mitigating heap fragmentation and toolchain limitations. Optimized data sorting using pointer-array manipulation.
- **Fail-Safe Data Serialization & Rollback:** Engineered an instant disaster-recovery subsystem to secure device state during unexpected field power blackouts. Leveraged a continuous pool-memory model to execute direct binary dumps of active class states onto an MMC storage device over SPI using a customized **miniFAT** file system layer, enabling immediate atomic rollbacks.
- **Niskopoziomowy Debugging & Async Interrupt Architecture:** Diagnosed and resolved a critical 3-month elusive system crash (Heisenbug) caused by IRQ stack pointer misalignment following an LCD hardware upgrade. Restructured the codebase into a robust **Top-Half** / **Bottom-Half** paradigm using functional pointers to decouple interrupt vectors from processing logic.
- **Radio Telemetry & RFID Integration:** Integrated a hybrid multi-channel data acquisition matrix. Authored low-level configuration matrices for RFID receiver registers (anticollision filters, RSSI thresholds) and implemented an active, real-time data-alignment algorithm pairing asynchronous RFID tag strings with synchronous microsecond hardware pulses.
- **Asynchronous HMI & Custom Graphics Engine:** Designed a non-blocking execution pipeline managing simultaneous matrix keypad polling (software debouncing), character/OLED graphics arrays, and high-current external stadium LED displays. Programmed a rasterization and dynamic scaling algorithm for bitmap fonts optimized for extreme sub-zero readability.
- **Desktop C++/MFC Architecture & B2B API Design:** Built a Windows desktop command center application in Visual Studio using **C++/MFC** adhering to the Document/View architecture. Developed a high-precision clock synchronization protocol to actively offset crystal drift and Round-Trip Time latency. Authored a custom JSON parser to handle automated B2B telemetry sync with national registration platforms.

Lead E-commerce Architect & Growth Analytics Specialist **2015 – 2025**

Sarenka (Digital Retail and E-commerce Ecosystems)

- **E-commerce Architecture & Traffic Acquisition:** Orchestrated the design, structural technical optimization, and high-performance Search Engine Optimization (SEO) of corporate web stores to maximize visibility across global search engines.
- **Conversion Rate Optimization & Funnel Analysis:** Architected advanced, multi-tier transactional funnels and mapped user behavior through **Google Analytics**. Executed programmatic UI/UX data-driven shifts based on empirical conversion data.
- **Performance Campaign Engineering:** Engineered, deployed, and systematically managed large-scale premium ad budgets inside **Google AdWords (Ads)**, delivering sustained increments in Return on Ad Spend (ROAS).

Independent Embedded Systems & Smart Home Automation Engineer **2016 – Present**

- **Nice TTBus Industrial Reverse Engineering:** Executed deep protocol analysis and reverse engineering of the closed-source, industrial **Nice TTBus** physical and packet layers.
- **ARM Hardware R&D:** Designed and fabricated an independent physical ARM-based gateway controller bridging Nice TTBus networks with modern Building Management Systems (BMS), preventing expensive motor replacement overhauls.

Akademia Sarenka Software Kids (APEX System) **2026 – Present**

Role: Chief Systems Architect & Co-Founder

- **Franchise Core Architecture (SaaS/MaaS):** Engineered a turnkey, zero-overhead automated business architecture ("Business-in-a-Box") designed for nationwide franchise replication in premium digital youth education.
- **Automated Invoicing Data Pipelines:** Integrated asynchronous landing page front-ends with ActiveNow membership systems and iFirma financial APIs, constructing automated month-end billing pipelines with zero fixed cost overhead.
- **Pre-Project Agile Dydaktyka:** Authored a highly effective 31-week curriculum introducing young dewelopers to advanced GitHub Projects tracking workflows, establishing project management discipline prior to writing text code.

🎓 EDUCATION, DIPLOMAS & CERTIFICATES

TELC English B2 International Language Certificate **Issued: 06/2014**

Centrum Nauki W&W (Jelenia Góra, Poland) — Common European Framework (CEFR) Fluency Qualification

Master of Science (M.Sc. Eng.) in Electronics & Telecommunications **2003 – 2005**

Wrocław University of Science and Technology (WUST) — Main Campus, Wrocław

Degree Level: Graduate M.Sc. Studies (Uzupełniające Magisterskie) | Specialization: ***Information Systems***

- **Master's Thesis (Rector's Honors, Grade: Excellent / Celująca):** *"Design and Implementation of an Interactive Real-Time Call Center System"*. Conceived and fully programmed an advanced multi-threaded voice platform based on early Voice-AI technology (VoiceXML) featuring natural speech processing workflows. Engineered a high-throughput backend infrastructure in **Perl** coupled with a relational **MySQL** engine. Implemented robust transactional constraints (ACID) and explicit table/row-locking mechanisms to eliminate deadlock vectors under heavy, high-concurrency real-time telephonic traffic.

Bachelor of Science (B.Sc. Eng.) in Electronics & Telecommunications **1999 – 2003**

Wrocław University of Science and Technology (WUST) — Jelenia Góra Division

Degree Level: Undergraduate B.Sc. Studies (Inżynierskie) | Specialization: ***Telecommunication Networks***

- **Engineering Thesis:** *"Autonomous Decoder of Wireless Infrared Transmission Protocols"*. Designed and coded a low-level IR signal analyzer mapping non-standard protocols (RC5, RC6, NEC, Sony arrays) written entirely in raw **8051 Assembler**.
- **Rapid Prototyping Hardware Layout:** Independently routed the schematic topography and fabricated physical PCBs inside laboratory environments utilizing full photochemical procedures (transparency film alignment, UV photoresist exposure, chemical development, and acid copper etching). Performed direct hardware debugging of TSOP sensor outputs using logic analyzers to resolve time-domain jitter.

Electronics Technician Diploma (Tech. Eng.) **1994 – 1999**

Technical School of Electronics, Jelenia Góra, Poland

Specialization: ***General Electronics***

- **Graduate Specialization Project:** Conducted comprehensive laboratory diagnostics and theoretical processing drift validation of Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC), calculating quantization noise vectors and Nyquist sampling limit compliance.
- **Core Foundations:** Acquired extensive knowledge of digital and analog circuit topologies, component diagnostics, schematic reading, manual soldering, and early copper board chemical fabrication techniques.